MOS Transistor and CMOS fundamental

PRE-WORK

SEMICONDUCTOR IC DESIGN FOUNDATION

AND READINESS PROGRAMME

2-Hour Training Module

Module 4 at a glance

ORIENTATION

Introduce participants to the semiconductor device fundamentals behind modern integrated circuits. Participants will understand how MOSFETs operate as electronic switches and how NMOS and PMOS transistors are used to construct CMOS digital circuits.

Learning outcomes

ORIENTATION

Basic structure of NMOS and PMOS transistors; gate, source and drain terminals; transistor switching operation; ON and OFF states; basic CMOS operation; how NMOS and PMOS transistors form logic gates; voltage, current and resistance fundamentals; threshold voltage; relationship between transistor behaviour, power consumption and circuit speed.

MOS Transistor

❷ Change the current layer into N+ diffusion by a click on the palette of the Diffusion N+ button. Make sure that the red layer is now the N+ Diffusion. Draw a n-diffusion box at the bottom of the drawing as in Figure 3-4. N-diffusion boxes are represented in green. The intersection between diffusion and polysilicon creates the channel of the nMOS device.

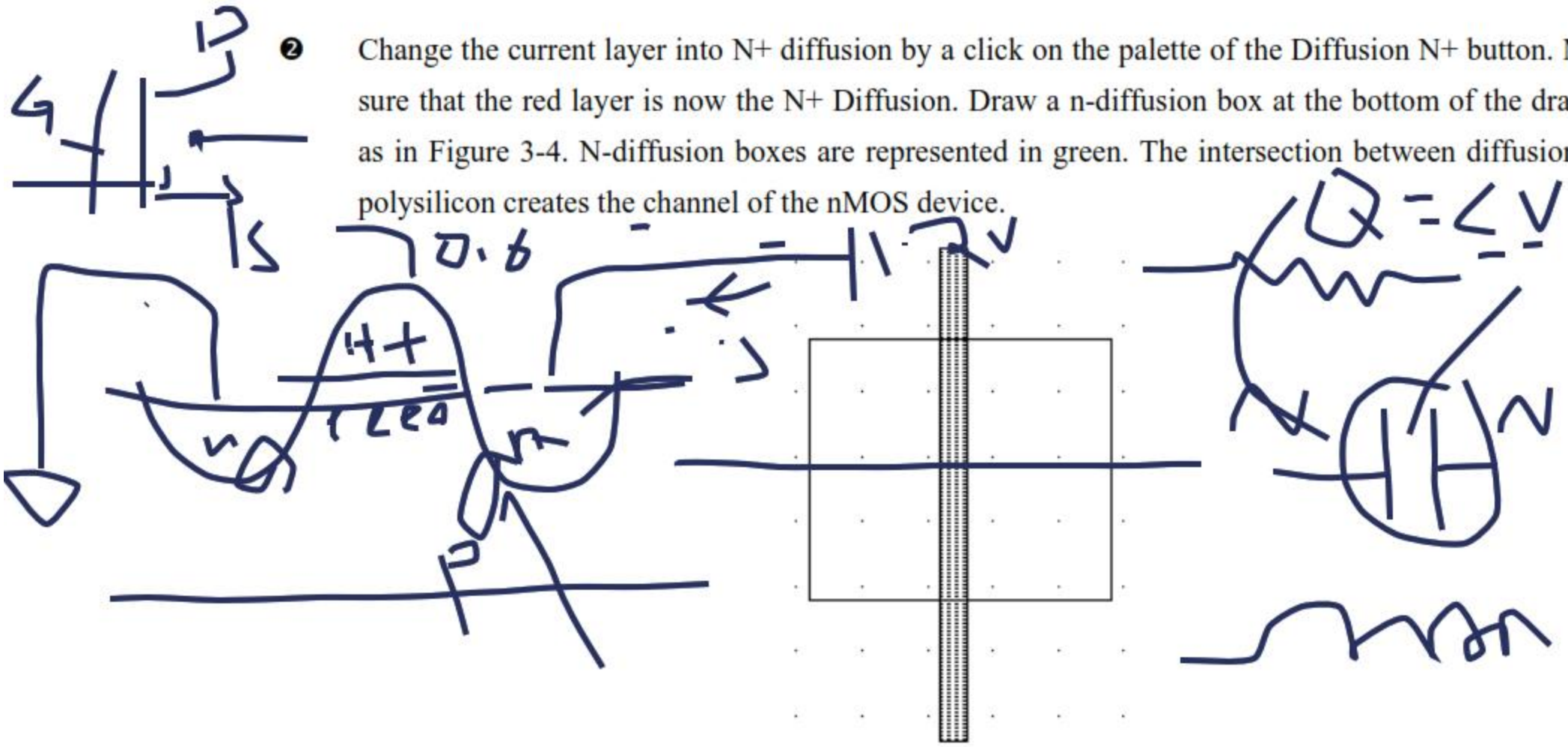


Fig. 3-4. Creating the N-channel MOS transistor

MOS Transistor

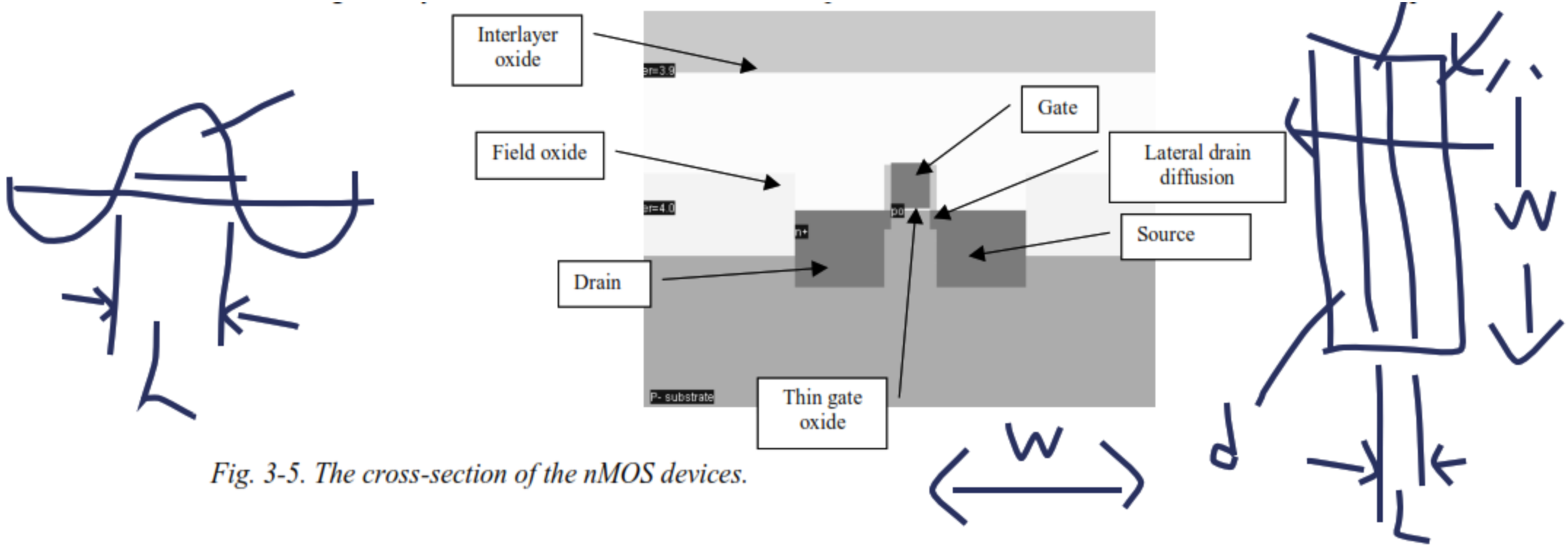
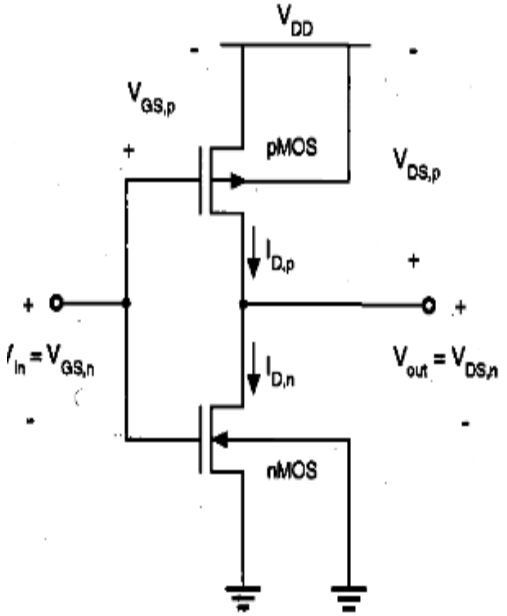
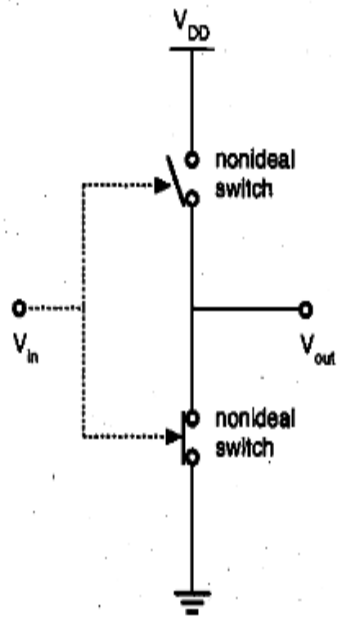
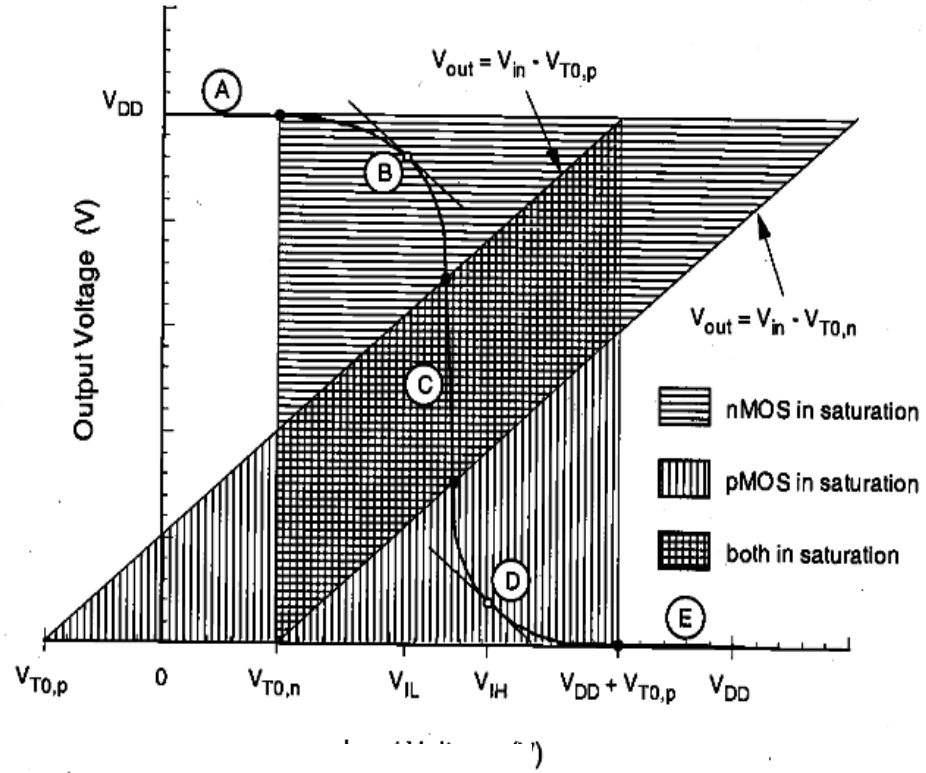


Fig. 3-5. The cross-section of the nMOS devices.

CMOS Inverter





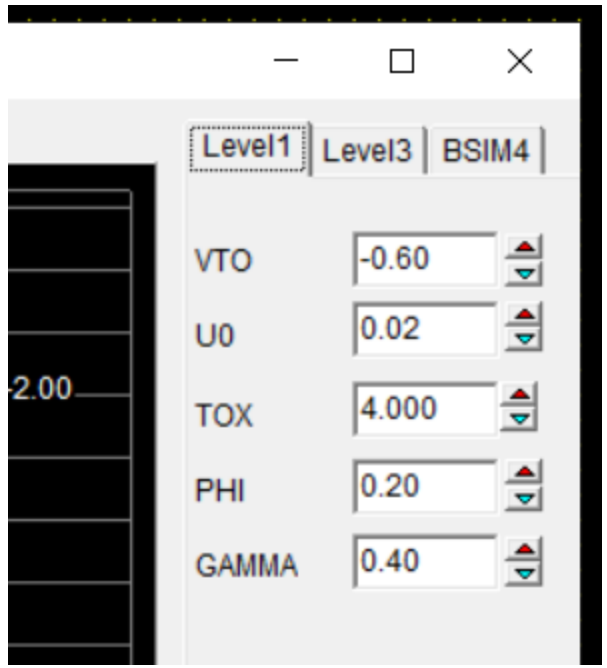


Region	V_{in}	V_{out}	nMOS	pMOS
A	$< V_{T0,n}$	V_{OH}	cut-off	linear
B	V_{IL}	high $\approx V_{OH}$	saturation	linear
C	V_{th}	V_{th}	saturation	saturation
D	V_{IH}	low $\approx V_{OL}$	linear	saturation
E	$> (V_{DD} + V_{T0,p})$	V_{OL}	linear	cut-off

MOS Transistor Model

Ref: cmos018.rul

Parameter	Keyword	Definition
VTO	l3vto	Threshold voltage
U0	l3u0	Low field mobility
PHI	l3phi	Surface potential at strong inversion
LD	l3ld	Lateral diffusion into channel
GAMMA	l3gamma	Bulk threshold parameter
KAPPA	l3kappa	Saturation field factor
VMAX	l3vmax	Maximum drift velocity
THETA	l3theta	Mobility degradation factor
NSS	l3nss	Sub-threshold factor
TOX	l3tox	Gate oxide thickness



Example Inverter Simulation

the charge or discharge of the output node. Without any switching activity, the current is almost equal to zero.

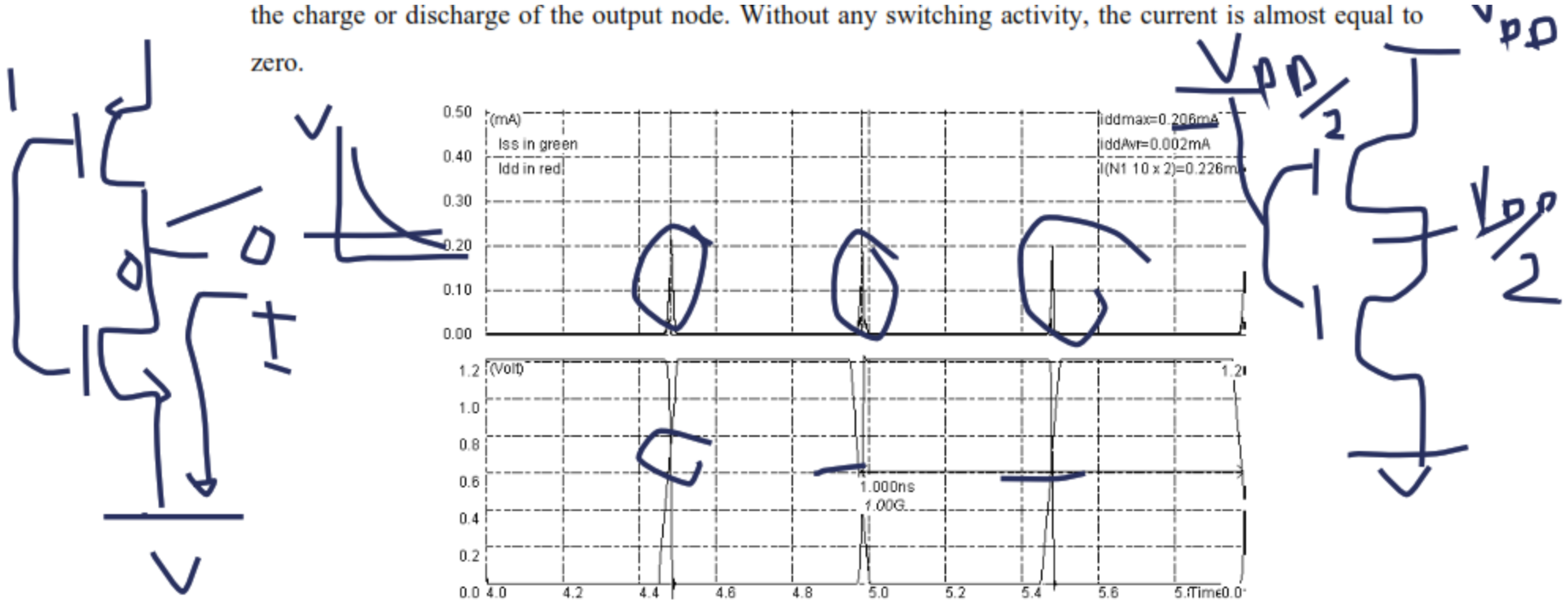


Fig. 4-14: Simulation of the current peaks appearing between VDD and VSS in the CMOS inverter at each output transition (InvSteps.MSK)

Layout – click at the layout



Capacitance at 'A'

Signal path at input 'A'

Power

$$P = \frac{1}{2}\,\eta . C\, V_{DD}^{\,2}\, f$$

Where:

k: technological factor (close to 1)

C: Output load capacitance (Farad)

VDD: supply voltage (V)

f: Clock frequency (Hz)

η: switching activity factor (Between 0 and 1)



Group Exercise

Task

1. Choose only one digital circuit; identify the digital gates from the verilog code (if you use it to compile for layout)
2. Show the signal path of the circuit (Show at layout)
3. Explain the delay based on the timing diagram.
4. Determine the current or power consume by the circuit
5. Repeat (3) and (4) for 180nm (cmos018) and 50 nm (cmos50n) processes.
6. Calculate the power based on the total capacitance and clock frequencies.

Knowledge check

WRAP-UP

1. Device physics can be expressed by equations.

2. The power consumption occurs briefly during transitions of the output, either from 0 to 1 or from 1 to 0. The current consumption is important only during a very short period corresponding to the charge or discharge of the output node.

3. Equation constant can be validated or derived from measured/simulate data.

Module 4: submit report
https://drive.google.com/drive/folders/1pJiXCG217wpYqlUotU7CZ2s9T534I9iQ?usp=drive_link